

Responsible & Ethical AI

Complete Tutorial: Ethical Frameworks, Principles, Bias, Governance, Regulation

GARP RAI Exam Preparation

Slide 1: What is Responsible AI?

Definition of Responsible AI:

- Responsible AI refers to the ethical and responsible development, deployment, and use of artificial intelligence systems
- Aims to ensure that AI benefits society holistically while mitigating potential risks
- Involves aligning AI systems with ethical standards, industry standards, regulation, legislation, and global principles
- Ensures AI operates in a fair, transparent, and accountable manner

Key Components:

- 1 **Ethical Standards:** Aligning with moral principles and values
- 2 **Industry Standards:** Following best practices and guidelines
- 3 **Regulation and Legislation:** Complying with laws and regulations
- 4 **Global Principles:** Adhering to international norms and agreements

Key Insight: Responsible AI is about **ethical development and deployment** of AI systems!

Why Ethical Frameworks Matter in AI:

- Ethical frameworks help us decide how to design AI and how to deploy it
- They help minimize risks, harms, and wrongs
- They shape AI by helping us define principles to abide by
- They help identify unacceptable biases
- They assess what kind of transparency is desirable
- They identify stakeholders and their interests
- They promote accountability
- They make justifiable decisions about design and implementation

Key Benefit: Different ethical frameworks help us make justifiable decisions in pluralistic societies where people hold diverse values.

Key Insight: Ethical frameworks [guide decision-making](#) to minimize risks and harms!

What is Practical Ethics?

- The application of ethical theory and the development of good practices to solve real-world moral dilemmas
- Provides concrete guidance for moral decision making and problem solving
- Goes beyond theoretical philosophy to offer actionable advice

Key Characteristics:

1 Applied Focus:

- Addresses actual ethical dilemmas
- Offers recommendations, not just theories
- Domains: medicine, law, politics, business, AI

2 Multidisciplinary:

- Draws on moral philosophy
- Social sciences (psychology, sociology)
- Domain expertise, law, computer science

3 Context-Sensitive:

- Situational nuances matter
- One-size-fits-all approaches don't work

4 Pluralistic:

- Blends different ethical frameworks
- Aims for justifiable decisions

Key Insight: Practical ethics = **applying theory to real-world dilemmas!**

Slide 4: Why Firms Consider Ethics Frameworks

Business Benefits of Practical Ethics:

Building Trust and Reputation:

- Customers, employees, and the public trust ethical companies
- Improves brand image
- Creates competitive advantage

Avoiding Scandals:

- Unethical behavior damages reputation
- Fraud, bias, privacy violations can be catastrophic
- Ethics helps prevent major scandals

Attracting and Retaining Talent:

- Workers increasingly care about ethical practices
- Ethics frameworks help recruit and retain talent
- Values-aligned workforce

Key Insight: Ethics makes [good business sense](#) with multiple benefits!

Slide 5: Ethics and Law Relationship

How Ethics and Law Relate:

Ethics as Complement to Law:

- Ethics helps ground, inform, and shape law
- Societies regulate behavior based on moral acceptability
- Ethics helps distinguish between just and unjust laws

Laws are Narrow in Scope:

- Establish minimal behavioral requirements
- Enable orderly interactions within basic fairness
- Provide a framework for society

Ethics Goes Beyond Law:

- Identifies moral issues
- Reflects on the kind of society we want
- Makes recommendations for a good life
- More ambitious than law

Example:

- It's not illegal to abandon friends, but it's immoral
- Laws provide minimums; ethics provides ideals

Key Insight: Ethics **complements and informs** law - goes beyond minimum requirements!

Slide 6: Strengthening Organizational Culture

How Ethics Strengthens Culture:

Key Cultural Benefits:

1 Teamwork:

- Ethics programs nurture collaboration
- Shared values create alignment
- Better team dynamics

2 Accountability:

- Clear ethical standards
- Responsibility for actions
- Transparent decision-making

3 Integrity:

- Doing the right thing
- Ethical leadership
- Trustworthy practices

Results:

- Boosts morale
- Increases productivity
- Creates positive work environment
- Reduces ethical lapses

Key Insight: Ethics frameworks strengthen **teamwork**, **accountability**, and **integrity**!

7: Supporting Risk Management

Risk Management Benefits of Ethics Frameworks:

Key Risk Reductions:

1 Safety Failures:

- Ethical design considers safety
- Proactive risk identification
- Prevention of harmful outcomes

2 Data Breaches:

- Privacy as ethical priority
- Strong security measures
- Data minimization

3 Regulatory Non-Compliance:

- Ethics frameworks anticipate regulation
- Proactive compliance
- Avoid penalties

How It Works:

- Ethical frameworks undergird policies and procedures
- Provide guidance for difficult decisions
- Create accountability structures

Key Insight: Ethics frameworks **reduce risks** of failures, breaches, and non-compliance!

Slide 8: Guiding Responsible Innovation

How Ethics Guides Innovation:

Responsible Innovation:

1 Consider Impacts:

- Ethics ensures consideration of people and society
- Balances progress with protection
- Prevents harmful applications

2 Direct Development:

- Ethics provides direction for innovation
- Identifies acceptable boundaries
- Encourages beneficial applications

3 Build Trust:

- Ethical innovation builds public trust
- Encourages adoption
- Creates sustainable progress

Key Principle:

- Innovation should benefit society
- Ethics ensures technology serves human values
- Prevents technology from serving only narrow interests

Key Insight: Ethics guides innovation in **responsible directions** considering societal impacts!

Slide 9: Promoting Long-Term Thinking

How Ethics Promotes Long-Term Perspectives:

Long-Term Benefits:

1 Sustainability:

- Considering future impacts
- Environmental and social sustainability
- Intergenerational justice

2 Stakeholder Value:

- Balancing short and long-term interests
- Considering all affected parties
- Building lasting relationships

3 Resilience:

- Ethical companies weather crises better
- Trust provides buffer
- Long-term viability

Key Insight:

- Managing with ethical considerations promotes long-term view
- Counteracts short-term profit pressures
- Builds enduring value

Key Insight: Ethics promotes **long-term thinking** and sustainability!

The Business Case for Practical Ethics:

Why Ethics Makes Business Sense:

1 Trust and Reputation:

- Ethical companies are trusted
- Better brand image
- Customer loyalty

2 Risk Reduction:

- Avoid scandals
- Prevent legal issues
- Reduce regulatory exposure

3 Talent Acquisition:

- Attract values-aligned employees
- Retain skilled workers
- Build committed teams

4 Culture:

- Stronger workplace culture
- Higher productivity
- Better collaboration

Key Point: Proponents argue that practical ethics is not just morally right but also financially prudent.

Key Insight: Ethics makes **good business sense** with measurable benefits!

How Ethics Manifests in Organizations:

Practical Applications:

1 Decision-Making:

- Ethics provides frameworks for difficult choices
- Guides trade-off decisions
- Ensures consistency

2 Policy Development:

- Ethics undergirds policies
- Creates accountability structures
- Establishes standards

3 Training:

- Ethics training for employees
- Awareness of ethical issues
- Skills for ethical decision-making

4 Oversight:

- Ethics committees
- Review processes
- Monitoring and enforcement

Key Insight: Ethics must be [embedded in organizational practices!](#)

Regulatory Recognition of Ethics:

Why Regulators Care About Ethics:

1 Protecting Citizens:

- Ethical standards protect individuals
- Prevent harm and discrimination
- Ensure accountability

2 Market Confidence:

- Ethical practices build trust
- Stable markets require trust
- Investor confidence

3 International Standards:

- Ethics frameworks guide regulation
- Global alignment
- Harmonized standards

Key Trend: Regulations increasingly require ethical considerations (e.g., GDPR, EU AI Act).

Key Insight: Ethics frameworks underlay **regulatory and legislative approaches!**

Slide 13: Ethics vs Compliance

Understanding Ethics vs Compliance:

Compliance:

- Following laws and regulations
- Minimum acceptable behavior
- External requirement
- Reactive approach
- "What must we do?"

Ethics:

- Going beyond legal requirements
- Aspirational standards
- Internal commitment
- Proactive approach
- "What should we do?"

Key Insight:

- Compliance is necessary but not sufficient
- Ethics provides the moral compass
- Ethics prevents issues that compliance doesn't address

Key Insight: Ethics goes **beyond compliance** - establishes aspirational standards!

Stakeholder Expectations for Ethical AI:

Key Stakeholder Groups:

1 Customers:

- Expect fair treatment
- Privacy protection
- Transparent decision-making

2 Employees:

- Ethical workplace
- Fair algorithms
- Meaningful work

3 Investors:

- Sustainable practices
- Risk management
- Long-term value

4 Regulators:

- Compliance
- Accountability
- Transparency

5 Society:

- Social good
- Fairness
- Trust

Key Insight: All stakeholders expect **ethical AI practices!**

Slide 15: Ethics Benefits Summary

Comprehensive Benefits of Practical Ethics:

Summary of Key Benefits:

1 Trust and Reputation:

- Enhanced brand image
- Customer loyalty
- Stakeholder confidence

2 Risk Management:

- Avoid scandals
- Reduce legal exposure
- Prevent regulatory issues

3 Talent Management:

- Attract top talent
- Retain employees
- Build engaged workforce

4 Culture:

- Stronger organizational culture
- Better teamwork
- Higher productivity

5 Innovation:

- Responsible innovation
- Sustainable development
- Long-term thinking

Key Insight: Ethics provides **multiple overlapping benefits** to organizations!

Three Major Ethical Frameworks:

1. Consequentialism:

- Judges morality based on consequences
- Most common: Utilitarianism
- Focus: Maximize overall utility
- Forward-looking, outcome-focused

2. Deontology:

- Judges morality based on duties and rules
- Key thinker: Immanuel Kant
- Focus: Adherence to ethical principles
- Duty-based, rule-focused

3. Virtue Ethics:

- Emphasizes virtuous character traits
- Key thinker: Aristotle
- Focus: Living a good life
- Character-based, holistic

Key Insight: These frameworks complement one another in practical ethics.

Key Insight: Three frameworks offer **different but complementary perspectives!**

Slide 17: Consequentialism Introduction

Consequentialism Explained:

Core Idea:

- Actions are judged by their consequences
- The morality of an action depends on its outcomes
- "The ends justify the means" (in some interpretations)

Key Principles:

1 Outcome Focus:

- What results from the action?
- Does it produce good or bad outcomes?
- Consequences determine moral value

2 Forward-Looking:

- Focus on future results
- Past actions judged by their effects
- Decisions based on expected outcomes

3 Circumstantially Relative:

- What's right depends on circumstances
- No absolute rules
- Context matters

Key Insight: Consequentialism = **judging actions by their consequences!**

Slide 18: Utilitarianism

Utilitarianism: The Most Common Form of Consequentialism

Core Idea:

- Maximize overall utility
- Utility often defined as happiness, pleasure, or satisfaction
- The morally correct action produces the greatest net utility

Key Thinkers:

1 Jeremy Bentham:

- Founder of utilitarianism
- Hedonic calculus
- Focus on pleasure and pain

2 John Stuart Mill:

- Refined utilitarianism
- Quality of pleasures
- Higher and lower pleasures

Application to AI:

- An AI system is morally good if it increases overall pleasure/happiness
- Actions are not intrinsically moral
- Derive moral value solely from results

Key Insight: Utilitarianism = maximize overall utility/happiness!

Slide 19: Consequentialism Advantages

Advantages of Consequentialism:

Key Advantages:

1 Single Quantifiable Metric:

- Provides one measure of moral value
- Utility can be calculated
- Clear decision-making framework

2 Impartiality:

- Everyone's utility counts equally
- No special treatment
- Objective calculation

3 Scientific Amenability:

- Utility can be measured
- Evidence-based decisions
- Quantifiable outcomes

4 Forward-Looking:

- Focuses on future outcomes
- Practical decision-making
- Real-world consequences

Note: Quantification of value can be challenging despite these advantages.

Key Insight: Consequentialism provides a [single quantifiable metric](#) for moral decisions!

Criticisms and Challenges of Consequentialism:

Key Challenges:

1 Subjectivity of Utility:

- Choice of metric is subjective
- Different definitions of "good"
- Measurement difficulties

2 Unpredictable Consequences:

- Can't predict all outcomes
- Where does the calculation stop?
- Immediate vs. long-term consequences

3 Rights Violations:

- Can justify violating individual rights
- "The ends justify the means"
- Classic example: Killing one to save five

4 Doctor Example:

- Killing a patient to use organs saves five
- Maximizes utility but seems immoral
- Challenge for consequentialism

Key Insight: Consequentialism can justify violating rights for the greater good!

Rule Consequentialism: A Modification

Core Idea:

- Judges acts by whether they adhere to utility-maximizing rules
- Not just by case-specific outcomes
- Addresses issues with act utilitarianism

How It Works:

1 Follow General Rules:

- Rules that tend to maximize utility
- Rules against murder, lying, etc.
- Upheld even when breaking them may increase utility

2 Example - Doctor Case:

- Doctor should not kill patient
- Would undermine trust in medical system
- Long-term utility is maximized by following rules

3 AI Application:

- Should not violate rights even if efficient
- Long-term trust is valuable
- Rule-based protection

Key Insight: Rule consequentialism = follow utility-maximizing rules!

Deontology Explained:

Core Idea:

- Judges morality based on adherence to duties and rules
- Not based on consequences
- Focus on moral obligations regardless of outcomes

Key Principles:

1 Duty-Based:

- Moral obligations are paramount
- We must do our duty
- Consequences don't override duties

2 Rule-Focused:

- Follow moral rules
- Universal principles
- Consistency across situations

3 Respect for Persons:

- People are ends in themselves
- Never treat as mere means
- Intrinsic value of individuals

Key Insight: Deontology = judging actions by duties and rules!

Kant's Contribution to Deontology:

Key Thinker:

- Immanuel Kant is the most important proponent of deontology
- Emphasized moral principles, rationality, and duty
- Famous for not placing moral weight on consequences
- Believed lying is always wrong regardless of consequences

Core Concepts:

1 Categorical Imperative:

- Act only according to maxims that could be universal law
- "It's wrong to lie" - applicable universally
- No contradictions in universal application

2 Treating People as Ends:

- Never treat people as mere means to an end
- People have intrinsic value
- Respect their autonomy and dignity

3 Reason-Based Morality:

- Morality grounded in reason
- Not in emotions or consequences
- Rational beings can determine moral duties

Key Insight: Kant = duty-based ethics focused on universal principles!

Understanding the Categorical Imperative:

First Formulation:

- "Act only according to that maxim whereby you can at the same time will that it should become a universal law"
- Can the rule be applied universally without contradiction?
- Example: Could "It's okay to lie" be a universal law?

Second Formulation:

- "Act in such a way that you treat humanity, whether in your own person or in the person of any other, always at the same time as an end and never merely as a means"
- Respect the dignity of all persons
- Don't use people as tools

Application to AI:

- AI systems should respect human dignity
- People should not be treated as data points
- Transparency and consent are essential

Key Insight: Categorical imperative = **act on rules that could be universal law!**

Contemporary Deontological Ethics:

Evolution of Deontology:

- Most contemporary deontologists care about consequences
- However, consequences are not the only moral consideration
- There are red lines that should not be crossed

Key Features:

1 Rights:

- Individuals have fundamental rights
- Rights cannot be violated even for good outcomes
- Examples: Privacy, autonomy, non-discrimination

2 Rules:

- Established ethical rules matter
- Professional codes of conduct
- Organizational policies

3 Principles:

- Ethical principles provide guidance
- Beyond consequences
- Deontological in nature

Key Insight: Modern deontology considers consequences but has red lines!

Criticisms and Challenges of Deontology:

Key Challenges:

1 Rigidity:

- Can lead to rigid rule-following
- Even when consequences are disastrous
- Example: Kant said lying is always wrong

2 Conflicting Duties:

- What if duties conflict?
- Example: Duty to tell truth vs. duty to protect
- No clear resolution framework

3 Application Difficulty:

- Hard to apply in complex situations
- What rules apply?
- How to resolve conflicts?

AI Application:

- Can lead to overly rigid AI systems
- May not adapt to context
- Difficulty encoding all rules

Key Insight: Deontology can be **too rigid** even when consequences are disastrous!

Virtue Ethics Explained:

Core Idea:

- Emphasizes virtuous character traits
- Focus on living a good life
- Rather than rules or consequences

Key Principles:

1 Character Focus:

- What kind of person should I be?
- Cultivating virtues
- Becoming a good person

2 Virtues:

- Wisdom, courage, humanity
- Justice, temperance, generosity
- Character traits to cultivate

3 Practical Wisdom:

- Exercising judgment in situations
- Moderating emotions and appetites
- Appropriate behavior in context

Key Insight: Virtue ethics = focus on character and living a good life!

Aristotle's Contribution:

Historical Roots:

- Virtue ethics has roots in ancient Greek philosophers
- Aristotle taught that happiness comes from a life guided by virtues
- Virtues are nurtured through practice and habit

Key Concepts:

1 Eudaimonia:

- Flourishing or living well
- Ultimate goal of human life
- Achieved through virtue

2 Virtues as Habits:

- Virtues are developed through practice
- Repeated actions create character
- Modeling virtuous exemplars

3 Golden Mean:

- Virtue lies between extremes
- Courage between cowardice and recklessness
- Balance in character

Key Insight: Aristotle = virtues through practice for a good life!

Slide 29: Key Virtues

Examples of Virtues:

Core Virtues:

1 Wisdom:

- Practical judgment
- Understanding what matters
- Sound decision-making

2 Courage:

- Facing challenges appropriately
- Moral courage
- Standing up for what's right

3 Humanity:

- Compassion and care
- Empathy for others
- Benevolence

4 Justice:

- Fairness and equity
- Giving proper due
- Social responsibility

5 Temperance:

- Self-control
- Moderation
- Balance

6 Generosity:

- Giving freely
- Sharing resources
- Beneficence

Key Insight: Virtues include [wisdom](#), [courage](#), [justice](#), [temperance](#), and [generosity](#)!

Applying Virtue Ethics:

Practical Approach:

1 Ask the Virtue Question:

- "What would a virtuous person do?"
- Consider role models and exemplars
- What would Jesus, Mohammed, Gandhi do?

2 Cultivate Virtues:

- Through practice and habit
- Learning from experience
- Modeling virtuous behavior

3 Consider Character:

- Not just discrete acts
- Life-long character development
- Holistic approach to morality

Application to AI:

- Build AI that learns from experience like children
- Develop virtuous organizational culture
- Consider what responsible companies would do

Key Insight: Virtue ethics asks "What would a virtuous person do?"!

Criticisms of Virtue Ethics:

Key Challenges:

❶ Lack of Clear Guidance:

- Less concrete than rules-based approaches
- Hard to know what virtue requires
- Situational judgment needed

❷ Conflicting Virtues:

- Different virtues can conflict
- How to weigh courage vs. prudence?
- No clear resolution method

❸ Cultural Relativism:

- Virtues vary across cultures
- What's virtuous in one context may not be in another
- Need for universal standards

Virtue Ethicist Response:

- Practical wisdom helps navigate hard cases
- Other theories also have conflicts
- Consequences not always comparable

Key Insight: Virtue ethics criticized for **lack of clear guidance!**

Contemporary Applications of Virtue Ethics:

Extension to Organizations:

- Virtue ethics expands beyond individual character
- Organizational virtues include:
 - Justice
 - Accountability
 - Environmental stewardship
 - Responsible innovation

Renewed Interest:

- Across moral psychology
- Business ethics
- AI ethics and development

Integration with Common Morality:

- People learn morality through habituation
- Parents teach children through repeated examples
- Virtue ethics aligns with how morality develops

Key Insight: Organizational virtues include [justice](#), [accountability](#), and [responsible innovation](#)!

Applying Virtue Ethics to AI:

Virtue-Based AI Development:

1 Learning from Experience:

- Some scholars suggest AI should learn like children
- Through experience and habit
- Not just coded rules

2 Organizational Virtues:

- Responsible development
- Ethical corporate culture
- Virtuous design teams

3 Character of AI:

- What would an ethical AI do?
- Embedding virtues in AI systems
- Training on ethical examples

Key Insight:

- Morality is complex - may be impossible to code top-down
- Learning approach may be more effective
- Ethics as character development

Key Insight: Virtue ethics AI would [learn from experience like children!](#)

How Frameworks Complement Each Other:

Combined Approach:

- Consequentialism, deontology, and virtue ethics are not mutually exclusive
- In practical ethics, they complement one another
- The best moral decision accords with all three

Integrated Decision-Making:

1 Consequentialist Analysis:

- What are the likely outcomes?
- Does it maximize good consequences?
- Consider all affected parties

2 Deontological Analysis:

- Does it respect rights and duties?
- Are there red lines being crossed?
- Is it in accordance with principles?

3 Virtue Ethics Analysis:

- What would a virtuous person do?
- Does it build character?
- Is it consistent with values?

Key Insight: Frameworks are **complementary**, not mutually exclusive!

Slide 35: Ethical Frameworks Summary

Summary of Ethical Frameworks:

Consequentialism:

- Focus: Outcomes and consequences
- Key Question: "What maximizes good outcomes?"
- Strength: Single quantifiable metric
- Weakness: Can justify rights violations

Deontology:

- Focus: Duties and rules
- Key Question: "What are our moral obligations?"
- Strength: Respects rights and dignity
- Weakness: Can be rigid

Virtue Ethics:

- Focus: Character and virtues
- Key Question: "What kind of person should I be?"
- Strength: Holistic and aspirational
- Weakness: Lack of clear guidance

Key Insight:

- Each framework offers valuable insights
- Best decisions often integrate all three
- Context determines emphasis

Key Insight: Three frameworks provide [complementary perspectives](#) for ethical AI!

Medical Ethics: A Historical Overview

Ancient Foundations:

- Hippocratic Oath (5th-3rd century BC)
- "Do no harm" to patients
- First known code of medical ethics

Key Documents:

① Nuremberg Code (1947):

- Response to Nazi medical experiments
- Informed consent requirement
- Research ethics foundation

② Declaration of Geneva (1948):

- Modern Hippocratic Oath
- International medical ethics

③ Declaration of Helsinki (1964):

- Research ethics guidelines
- Updated regularly
- Global standards

④ Belmont Report (1978):

- Respect for persons, beneficence, justice
- Research ethics framework
- US foundational document

Key Insight: Medical ethics has [long history](#) and [established frameworks](#)!

Medical Ethics Development in the 1970s:

Factors Driving Evolution:

1 Concentration of Medical Care:

- Hospitals and depersonalized settings
- Loss of personal relationships
- Need for formal ethics

2 Rising Costs and Government Role:

- Increased government involvement
- Healthcare funding changes
- Resource allocation decisions

3 Patient Rights Movements:

- Civil rights outgrowth
- Demands for autonomy
- Informed consent

4 Medical Scandals:

- Tuskegee Syphilis Experiment
- Public outrage
- Call for reform

5 Rapid Technological Advances:

- Mechanical ventilators
- Organ transplantation
- New ethical dilemmas

Key Insight: Multiple factors drove [development of medical ethics](#)!

Technology Created New Ethical Challenges:

Example - Mechanical Ventilator:

- Prompted reconsideration of death
- Heart-beating bodies with non-functioning brains
- Organ procurement dilemmas
- Is the patient alive or dead?

Key Lesson:

- Technological advances create ethical challenges
- Moral questions, not medical ones
- Need for ethical frameworks

Parallels to AI:

- AI creates new ethical dilemmas
- Not purely technical questions
- Need for ethical guidance

Key Insight: Technology creates [new ethical dilemmas](#) that need frameworks!

What AI Ethics Can Learn from Medical Ethics:

Key Lessons:

1 Structured Frameworks:

- Medical ethics has established principles
- AI needs similar foundations
- Clear ethical guidance

2 Professional Ethics:

- Doctors have ethical training
- AI professionals need ethics education
- Professional responsibility

3 Ethics Committees:

- Medical ethics committees
- AI needs similar oversight
- Multi-disciplinary review

4 Codes of Ethics:

- Medical codes are well-established
- AI codes are evolving
- Need for standards

Key Insight: Medical ethics provides [model for AI ethics development!](#)

Parallels Between Medical and AI Ethics:

Similar Challenges:

- ❶ **High-Stakes Decisions:**
 - Medicine: Life and death
 - AI: Criminal justice, finance, healthcare
- ❷ **Rapid Technological Change:**
 - Medicine: Mechanical ventilation, transplants
 - AI: Rapid advancement, unpredictable effects
- ❸ **Need for Oversight:**
 - Medicine: Ethics committees
 - AI: Governance structures needed
- ❹ **Professional Responsibility:**
 - Medicine: Doctors' oaths
 - AI: Developers' responsibilities

Key Difference:

- Medical ethics has centuries of development
- AI ethics is still evolving
- Can learn from medical ethics

Key Insight: AI ethics can [learn from established medical ethics](#) structures!

Five Core Principles of AI Ethics:

The Five Principles:

1 Nonmaleficence:

- Obligation to avoid harming others
- "Do no harm"
- Prohibition of unjustified risks

2 Beneficence:

- Obligation to act for the benefit of others
- "Do good"
- Active help and support

3 Justice:

- Fairness, equality, impartiality
- Equitable distribution
- Non-discrimination

4 Autonomy:

- Respect for informed, un-coerced decisions
- Individual control
- Non-manipulation

5 Explainability:

- Transparent and understandable decisions
- Accountability
- Trust-building

Key Insight: Five principles = nonmaleficence, beneficence, justice, autonomy, explainability!

The Principle of Nonmaleficence:

Core Meaning:

- Obligation to avoid harming others
- Prohibition on imposing unjustified risks
- "First, do no harm" (primum non nocere)

Key Aspects:

1 Avoid Actual Harm:

- Don't directly harm others
- Example: Algorithms that discriminate
- Actions that cause injury

2 Avoid Unjustified Risk:

- Don't impose unnecessary risks
- Example: Untested algorithms
- Risks that outweigh benefits

3 Proportionality:

- Some risk is permissible
- If benefits justify risk
- If alternatives unavailable

Key Insight: Nonmaleficence = "do no harm" - avoid harming others!

When is Risk Acceptable?

Principles of Risk Acceptance:

1 Benefit Justification:

- Risk is acceptable if benefits justify it
- Example: Clinical trials
- Risk is part of medical progress

2 No Alternatives:

- Risk acceptable if no alternatives
- Example: Emergency procedures
- Only option available

3 Who Bears the Risk:

- More acceptable if those at risk benefit
- Example: Clinical trial participants
- Unacceptable if others bear risk

AI Application:

- Imposing algorithmic risk on people who don't benefit
- Unacceptable to experiment on populations
- Especially if they don't gain

Key Insight: Risk acceptable if **benefits justify it and risk-bearers benefit!**

The Principle of Beneficence:

Core Meaning:

- Moral obligation to act for the benefit of others
- Requires taking positive steps to help
- Moves beyond simply refraining from harm

Key Aspects:

1 Active Help:

- Donating to charity
- Volunteering
- Providing resources and assistance

2 Beyond Nonmaleficence:

- Not just "do no harm"
- "Do good"
- Active beneficence

3 Deontological Duty:

- Duty-based obligation
- Not just about maximizing utility
- Required beyond consequences

Key Insight: Beneficence = "do good" - actively help others!

Limits to the Duty of Beneficence:

Reasonable Constraints:

1 Scarce Resources:

- Cannot help everyone
- Limited resources
- Prioritization needed

2 Competing Obligations:

- Multiple duties to balance
- Professional responsibilities
- Personal obligations

3 Reasonableness:

- There's only so much ethics can demand
- Demandingness constraint
- Proportionality matters

4 Recipient Autonomy:

- Unwanted "benefits" may be rejected
- Right to decide what's best
- Respect individual choices

Key Insight: Beneficence has [reasonable limits](#) - cannot help everyone!

Justice in the Context of AI:

Core Meaning:

- Fairness, equality, impartiality, proportionality
- Giving each person their proper due
- Upholding duties toward fairness and equality

Key Applications:

1 Fair Treatment:

- No discrimination based on protected characteristics
- Equal opportunities
- Impartial algorithms

2 Equitable Distribution:

- Benefits and burdens fairly allocated
- Access to AI benefits
- Protection from AI harms

3 Accountability:

- Those responsible should be accountable
- Just processes
- Fair consequences

Key Insight: Justice = fairness, equality, impartiality, and proportionality!

Four Types of Justice:

1. Procedural Justice:

- Fair processes and impartiality
- Right structures in place
- Rule of law

2. Distributive Justice:

- Equitable allocation of benefits and burdens
- Fair distribution of resources
- Access to opportunities

3. Restorative Justice:

- Repairing harms
- Reconciling victims and offenders
- Making amends

4. Interactional Justice:

- Respect and fairness between individuals
- Interpersonal treatment
- Dignity and respect

Key Insight: Four types of justice = procedural, distributive, restorative, interactional!

The Principle of Autonomy:

Core Meaning:

- Capacity to make informed, un-coerced decisions
- Respecting others' abilities to determine their own course
- Non-manipulation

Key Aspects:

1 Informed Decisions:

- People need adequate information
- Transparency about options
- Understanding consequences

2 Un-coerced Decisions:

- No coercion or manipulation
- Free choice
- Voluntary consent

3 Capacity:

- Must have capacity to decide
- Exceptions: children, unconscious patients
- Surrogate decision makers when needed

Key Insight: Autonomy = informed, un-coerced decisions!

Autonomy in Medical Context:

Informed Consent:

- Patients have right to voluntary informed consent
- Right to refuse treatment
- Doctor's duty to inform
- Patient's right to decide

What Undermines Autonomy:

1 Coercion:

- Forcing decisions
- Threats or pressure
- Removing choice

2 Deception:

- Lying or misleading
- Withholding information
- Manipulating understanding

3 Undue Influence:

- Excessive persuasion
- Exploitation
- Imbalance of power

Key Insight: Autonomy requires **informed consent and no coercion!**

Autonomy in the Context of AI:

Ethical AI and Autonomy:

- Ethical AI respects people's autonomy
- No coercive or manipulative tactics
- Technology should help people further their own goals
- Not serve third-party interests

Risks to Autonomy:

1 Manipulation:

- Recommendation algorithms
- Targeted content
- Behavioral influence

2 Coercion:

- Forced compliance
- No alternative
- System lock-in

3 Lack of Transparency:

- Users don't understand decisions
- Cannot make informed choices
- Autonomy undermined

Key Insight: Ethical AI respects **autonomy** by avoiding coercion and manipulation!

The Principle of Explainability:

Core Meaning:

- AI systems should provide transparent explanations
- Understandable reasons for actions or decisions
- Also called "explicability"

Why Explainability Matters:

1 Accountability:

- Hold designers and companies accountable
- Identify errors and biases
- Rectify unfair practices

2 Trust:

- Trust is fundamental for adoption
- Users trust understandable systems
- Transparency builds confidence

3 Ethical Compliance:

- Ensure adherence to ethical guidelines
- Respect individual rights
- Regulatory compliance

Key Insight: Explainability = transparent and understandable explanations!

Levels and Types of Explainability:

1. Local Explainability:

- Explains decisions on a single instance
- "Why was THIS decision made?"
- Example: Why was THIS loan denied?
- Specific to one prediction

2. Global Explainability:

- Explains overall model behavior
- "How does the model work generally?"
- Comprehensive understanding
- Across all inputs

3. Model-Specific Explainability:

- Techniques tailored to architecture
- Example: Decision tree rules
- Neural networks require different methods

4. Model-Agnostic Explainability:

- Works with any model
- Versatile and general
- LIME, SHAP are examples

Key Insight: Explainability has [different levels and types](#) for different needs!

Slide 53: Counterfactual Explanations

Counterfactual Explanations:

What They Are:

- Hypothetical scenarios contrasting with actual decisions
- Show conditions for different outcomes
- Provide actionable insights

Example - Loan Denial:

- Actual: Loan denied
- Counterfactual: "You would have been approved if:
 - You had \$10,000 more in the bank
 - OR you earned \$1,000 more per month

Benefits:

1 Transparency:

- Users understand decision factors
- See what influences outcomes

2 Actionable Insights:

- Know what to change
- Path to different outcome
- Empowerment

3 Fairness:

- Detect biases
- Understand disparate treatment
- Identify improvement areas

Key Insight: Counterfactual explanations show [what would lead to different outcomes!](#)

Challenges with Explainability:

Key Challenges:

1 What Counts as Explanation?

- No consensus on definition
- Different needs for different audiences
- Experts vs. general public

2 Who Gets Explanation?

- Affected individuals
- Regulators
- General public
- Different levels needed

3 Trade-offs:

- Accuracy vs. explainability
- Simple models are more explainable
- Complex models are more accurate

4 Technical Difficulty:

- Neural networks are "black boxes"
- Hard to trace decisions
- Technical solutions needed

Key Insight: Explainability faces **definitional**, **audience**, and **technical** challenges!

Summary of AI Ethics Principles:

Five Core Principles:

1 Nonmaleficence:

- "Do no harm"
- Avoid unjustified risks
- Proportional risk-benefit

2 Beneficence:

- "Do good"
- Active help
- Advance welfare

3 Justice:

- Fairness and equality
- Impartiality
- Non-discrimination

4 Autonomy:

- Informed decisions
- Non-coercion
- Respect for persons

5 Explainability:

- Transparency
- Understandable decisions
- Accountability

Key Insight: Five principles provide [comprehensive ethical framework](#) for AI!

Understanding Bias in AI Systems:

Definition:

- Systemic deviation in output or impact
- Compared to a desired norm or standard
- Algorithm is biased when it deviates from intended function

Example - Hiring Algorithm:

- Designed to identify best candidates
- But recommends based on race or sex
- Deviates from intended purpose
- This is problematic bias

Key Insight:

- No algorithm is perfectly objective
- Every algorithm reflects embedded values
- Values shape what's considered valuable

Key Insight: AI bias = systemic deviation from intended function!

When is Bias Acceptable?

Justifiable Bias:

- Part of well-designed AI
- Optimizes for relevant criteria
- Example: Loan eligibility prioritizing bank balance
- Relevant to loan repayment

Problematic Bias:

- Results in unfairness
- Disadvantages for unjustifiable reasons
- Based on irrelevant characteristics
- Example: Race or sex discrimination

Statistically Unbiased not equal to Ethically Acceptable:

- Even unbiased algorithms can cause harm
- Example: Charging exorbitant fees to everyone
- Statistically fair but ethically problematic

Key Insight: Not all biases are **problematic** - some are justifiable!

Four Primary Sources of Algorithmic Bias:

1. Problem Specification:

- Goals contain inherent problems
- Target variables miss real-world objectives
- Example: Healthcare expenditure proxy

2. Data:

- Historical bias
- Sampling bias
- Selective labels problem

3. Modeling, Validation, and Design:

- Optimization choices
- Model selection
- Subgroup consideration

4. Deployment:

- Human deference
- Unintended effects
- Context changes

Key Insight: Bias can enter at [any stage](#) of AI development!

Bias in Problem Specification:

How It Occurs:

1 Complex Goal Operationalization:

- Hard to capture real-world objectives
- Selected target variables may be poor proxies
- Example: Bank eliminating all risk

2 Unintended Consequences:

- Bank lends only to affluent
- Excludes others
- Fairness issue

3 Proxy Problems:

- Healthcare expenditure proxy
- Favored richest patients, not sickest
- Systematically biased

Key Lesson:

- Careful problem definition is essential
- Proxies must be thoughtfully chosen
- Goals should reflect true objectives

Key Insight: Problem specification can introduce **bias through poor operationalization!**

Biases in Data:

Historical Bias:

- Data reflects past discrimination
- Example: Women excluded from banking
- Algorithm favors men
- Perpetuates historical patterns

Selective Labels Problem:

- Missing counterfactual outcomes
- No data on rejected applicants
- Can't know if better candidates existed
- Perpetuates existing patterns

Sampling Bias:

- Non-random data sample
- Trends don't generalize
- Example: Data from young men only
- May not apply to women or elderly

Key Insight: Data biases include **historical**, **selective labels**, and **sampling bias**!

Bias in Modeling, Validation, and Design:

How Bias Emerges:

1 Optimization Functions:

- Choice of objective
- What is being optimized?
- May favor certain outcomes

2 Regression Models:

- Different models have different biases
- Choice matters
- Subgroup effects

3 Subgroup Consideration:

- How are subgroups handled?
- Are differences recognized?
- Fairness across groups

4 Information Presentation:

- How is data presented?
- Visual biases
- Interface effects

Key Insight: Modeling choices can [introduce bias](#) even with good data!

Search Engine Bias Example:

The Scenario:

- Search engine for financial products
- Designed to help users find best options
- Favors popular items
- Popularity leads to more exposure

The Bias:

- Perpetuates popularity cycles
- Quality may not matter
- Less popular but better options hidden
- Self-reinforcing bias

Lesson:

- Popularity not equal to quality
- Algorithmic bias can be subtle
- Design choices matter
- Fairness implications

Key Insight: Design choices can [create self-reinforcing biases!](#)

Bias in Deployment:

How It Occurs:

1 Human Deference:

- People defer to algorithm suggestions
- Even when limitations known
- Convenience factor

2 Reasons for Deference:

- Time-saving
- Perceived objectivity
- Self-doubt
- Responsibility shielding

3 Unintended Effects:

- Individuals relinquish responsibility
- Blame shifting
- Behavior changes

Key Lesson:

- Oversight is essential
- Human judgment remains important
- Don't blindly trust automation

Key Insight: Deployment can introduce [bias through human deference](#) to algorithms!

Why People Defer to Automation:

Convenience:

- Time-saving
- Effort reduction
- Easy to follow

Perceived Objectivity:

- Algorithms appear more objective
- People know their own fallibilities
- Creates self-doubt

Responsibility Shielding:

- Deferring shares responsibility
- Going against algorithm exposes people
- Blame avoidance

Consequences:

- Unintended effects
- Reduced oversight
- Bias perpetuation
- Accountability gaps

Key Insight: People defer to algorithms for convenience, objectivity perception, and responsibility shielding!

When Bias Becomes Discrimination:

Legal Perspective:

- Depends on jurisdiction
- When it disfavors based on protected characteristics
- Race, sex, ethnicity, age
- Other classifications protected by law

Protected Characteristics:

- 1 Race
- 2 Sex/Gender
- 3 Ethnicity
- 4 Age
- 5 Religion
- 6 Disability
- 7 Sexual orientation

Key Point:

- Such disadvantages typically violate legal protections
- Proactive measures needed
- Continuous monitoring

Key Insight: Discrimination = **disfavoring based on protected characteristics** in violation of law!

Understanding Fairness in AI:

Definition:

- Absence of bias or preference
- Based on irrelevant characteristics
- Algorithm is fair when it doesn't exhibit problematic biases

Two Types:

1 Group Fairness:

- Statistical criteria across groups
- Demographics mirror population
- Example: 51

2 Individual Fairness:

- Treating similar individuals similarly
- Challenge: Defining "similar"
- Focus on individual treatment

Key Insight: Fairness = [absence of problematic biases](#)!

Why Fairness Can't Be Automated:

Mathematical Impossibility:

- When base rates are unequal
- Cannot satisfy all fairness measures simultaneously
- Demographic parity, predictive rate parity, equal opportunity
- Trade-offs are unavoidable

Example - Criminal Risk Assessment:

- Men statistically more likely to commit certain crimes
- Algorithm assigns higher risk to man
- Man argues unfair treatment
- Woman argues unfair if assigned equal score

Key Point:

- Fairness is a moral judgment
- Not purely mathematical
- Compromises and trade-offs needed

Key Insight: Fairness cannot be automated when **base rates are unequal!**

Procedural Fairness in AI:

Definition:

- Fair processes and impartial procedures
- Not just fair outcomes
- Right structures in place

JUSTICE System Analogy:

- Procedural fairness: rule of law
- Outcome fairness: appropriate punishments
- Mistakes can be justifiable with fair processes
- Ways to right wrongs

AI Application:

- Ethics committees
- Established procedures
- Appeal processes
- Accountability structures

Key Insight: Procedural fairness = fair processes and impartial procedures!

Strategies to Avoid Problematic Biases:

Key Approaches:

1 Technological Solutions:

- Fairness toolkits (Aequitas, AIF360)
- Internal auditing systems
- Bias reports from providers
- But technology alone is insufficient

2 Data Quality:

- Diverse, updated, accurate data
- Representative samples
- Free from past discriminatory tendencies

3 Auditing:

- Algorithmic auditing is best approach
- Private companies offer services
- Independent assessment

Key Insight: Bias mitigation requires [multiple approaches beyond technology!](#)

Slide 70: Data Quality Importance

Data Quality and Bias Prevention:

Key Data Quality Factors:

1 Diversity:

- Include all relevant groups
- Avoid homogeneity
- Represent all populations

2 Accuracy:

- Correct and reliable data
- Avoid errors
- Verify sources

3 Representativeness:

- Reflect target population
- Avoid sampling bias
- Generalizable results

4 Currency:

- Up-to-date data
- Reflect current reality
- Avoid outdated patterns

Synthetic Data:

- Fabricated datasets
- No privacy risks
- Can be designed to avoid biases
- May be less precise

Key Insight: Data quality helps [avoid biases](#) through diversity and representativeness!

The Importance of Algorithmic Auditing:

Why Auditing Matters:

- Likely the best way to identify biases
- Fresh and diverse perspective
- Independent assessment
- Continuous improvement

What Auditing Includes:

1 Technological Tools:

- Statistical analyses
- Fairness metrics
- Bias detection

2 Human Review:

- Diverse perspectives
- Contextual understanding
- Qualitative assessment

3 Documentation:

- Findings and recommendations
- Remediation plans
- Continuous monitoring

Key Insight: Algorithmic auditing is **best way to identify and correct biases!**

The Role of Ethical Committees:

Purpose:

- Discuss possible problematic biases
- Make decisions about trade-offs
- Consider multiple ethical frameworks
- Ensure procedural fairness

Benefits:

1 Diverse Perspectives:

- Different viewpoints
- Multiple expertise areas
- Balanced considerations

2 Structured Decision-Making:

- Systematic review
- Transparent processes
- Documented reasoning

3 Accountability:

- Clear responsibility
- Audit trail
- Defensible decisions

Key Insight: Ethical committees help **discuss and address biases** through diverse perspectives!

Training for Leadership:

Why Leadership Education Matters:

- Wide range of risks from AI
- Financial, legal, reputational consequences
- Responsible AI is a strategic imperative
- Leadership drives organizational culture

Key Topics for Training:

1 AI Risks:

- Bias and discrimination
- Privacy violations
- Regulatory non-compliance
- Reputational damage

2 Ethical/Responsible AI Practices:

- Fairness principles
- Governance structures
- Risk mitigation

3 Strategic Considerations:

- Competitive advantage
- Trust building
- Long-term value

Key Insight: Leadership must be [educated about AI risks and responsible practices!](#)

Summary: Bias, Discrimination, and Fairness

Key Concepts:

- 1 **Bias:**
 - Systemic deviation from norm
 - Can be justifiable or problematic
 - Sources: Problem, data, modeling, deployment
- 2 **Discrimination:**
 - Bias that disfavors protected characteristics
 - Legally prohibited
 - Requires proactive measures
- 3 **Fairness:**
 - Absence of problematic bias
 - Group and individual fairness
 - Cannot be fully automated

Key Insight: Bias management requires [continuous effort across the AI lifecycle!](#)

Slide 75: Bias Mitigation Tools

Tools for Bias Mitigation:

Available Toolkits:

1 Aequitas:

- Tests models for different bias metrics
- Across different groups
- Auditing tool

2 AI Fairness 360 (AIF360):

- IBM's fairness toolkit
- Benchmark for testing
- Multiple fairness metrics

3 Internal Auditing Systems:

- Companies can create own systems
- Identify potential biases
- Continuous monitoring

External Options:

- Third-party auditing services
- Independent assessors
- Bias reports from providers
- Fresh perspective

Key Insight: Multiple tools available for **identifying and mitigating bias!**

Understanding Privacy in AI:

Definition:

- Privacy to extent others don't have access to personal information
- In reference to some person or institution
- In reference to some personal data point

Why Privacy Matters:

1 Protection from Abuse:

- More knowledge enables interference
- Power imbalances
- Control over individuals

2 Autonomy:

- Privacy supports self-determination
- Freedom from surveillance
- Personal space

3 Dignity:

- Respect for persons
- Not being reduced to data
- Human worth

Key Insight: Privacy = lack of access to personal information!

Understanding Data Security:

Definition:

- Doing what is necessary to prevent unauthorized access
- Protecting data from attacks and theft
- Ensuring data integrity

Key Components:

1 Physical Security:

- Hardware protection
- Storage device security
- Physical access controls

2 Administrative Controls:

- Access permissions
- Organizational policies
- Employee training

3 Software Applications:

- Encryption
- Authentication
- Monitoring systems

Key Insight: Data security = preventing unauthorized access to data!

Slide 78: Privacy as Ethical Issue

Why Privacy is an Ethical Issue:

Wrongs:

- Violates moral and legal rights
- Treats people instrumentally
- Reduces people to data points
- Respect for autonomy

Harms to Individuals:

- Discrimination
- Blackmail
- Exposure and shaming
- Identity theft

Harms to Institutions:

- Lawsuits and liability
- Fines and penalties
- Reputational damage

Harms to Society:

- National security risks
- Democratic threats
- Blackmail of officials

Key Insight: Privacy violations cause **wrongs and harms** to individuals, institutions, and society!

Slide 79: Personal Data as Toxic Asset

Personal Data: A Toxic Asset

Why "Toxic" ?

① Cheap to Mine:

- Easy to collect
- Low cost
- Widespread availability

② Valuable:

- Profitable for companies
- Marketing and targeting
- Business insights

③ Sensitive:

- Privacy concerns
- Vulnerable to abuse
- High risk

④ Hard to Keep Safe:

- Security challenges
- Attacker advantage
- Constant threats

Key Insight:

- Defenders are at disadvantage
- Attacker chooses time and method
- With enough resources, breach is inevitable

Key Insight: Personal data is a **toxic asset** - valuable but risky!

The Principle of Data Minimization:

Core Idea:

- Collect only necessary data
- Fulfill specific purpose
- Most effective privacy protection

Why It Matters:

1 Risk Reduction:

- Less data = less risk
- Fewer targets
- Reduced exposure

2 Respect for Persons:

- Privacy rights
- Avoid unnecessary intrusion
- Proportionality

3 Regulatory Compliance:

- GDPR requirement
- Privacy laws
- Best practice

Key Point:

- Only collect data when benefit outweighs disadvantages
- Data subjects deserve protection
- Minimal data collection is respectful

Key Insight: Data minimization = [collect only necessary data!](#)

The Right to be Forgotten:

Definition:

- Person's right to have private information removed
- From Internet searches and other directories
- When data is inadequate, irrelevant, or excessive

When It Applies:

1 Incorrect Information:

- Data that is wrong
- Inaccurate records
- False information

2 Irrelevant Information:

- No longer relevant
- Outdated
- No legitimate purpose

3 Excessive Information:

- More than needed
- Proportionality
- Over-collection

Key Insight: Right to be forgotten = removal of private information when inadequate or irrelevant!

Giving People Control Over Their Data:

Key Rights:

1 Consent Management:

- Easy to deny consent
- Easy to withdraw consent
- Clear options

2 Access:

- See personal data
- See inferences made
- Understand data use

3 Transfer:

- Data portability
- Move data to other services
- Freedom to choose

4 Deletion:

- Right to delete data
- Remove personal information
- Erasure

Key Insight: Control over data = consent, access, transfer, and deletion rights!

Contextual Integrity in Privacy:

Core Idea:

- Data use should adhere to contextual norms
- When data is transferred to different context, norms are violated
- Context matters

Examples:

1 Banking Context:

- Data given for transaction
- Should not be sold to marketing company
- Violates expectations

2 Healthcare Context:

- Patient data given to doctor
- Should not end up with data brokers
- Violates trust

3 Social Media Context:

- Expectations about data use
- Context-specific norms
- Violations cause backlash

Key Insight: Contextual integrity = data use should respect context-specific norms!

Best Practices for Data Deletion:

Why Delete Data:

1 Protect Individuals:

- Reduce exposure
- Privacy protection
- Security

2 Maintain Accuracy:

- Personal data changes quickly
- People change jobs, houses, preferences
- Outdated data is inaccurate

3 Reduce Liability:

- Less data = less risk
- Fewer targets
- Smaller breach impact

Best Practice:

- Regular deletion schedules
- Expiry dates for data
- Not collected with intention of keeping forever
- Routine data deletion

Key Insight: Delete data **as soon as no longer necessary** - routine practice!

Slide 85: Privacy Principles Summary

Summary of Privacy Principles:

Key Principles:

1 Right to Privacy:

- Moral and legal right
- Universal Declaration of Human Rights
- Constitutional protections

2 Data Minimization:

- Collect only necessary data
- Most effective protection

3 Right to be Forgotten:

- Remove inadequate data
- Irrelevant or excessive

4 Control Over Data:

- Consent, access, transfer, deletion

5 Contextual Integrity:

- Respect context-specific norms

6 Data Deletion:

- Delete when no longer necessary

7 Data Security:

- Encryption, anonymization, differential privacy

Key Insight: Multiple principles guide [ethical privacy practices](#)!

Why AI is Difficult to Govern:

Key Challenges:

1 Power Asymmetries:

- Tech companies more powerful than governments
- Wealth, influence, expertise
- Regulation difficulties

2 Lack of Transparency:

- Private company development
- Limited public information
- Safety testing unknown

3 Lack of Interpretability:

- Neural networks are "black boxes"
- Proxy variables obscure
- Hard to audit

4 Lack of Ethics Structures:

- No required ethics education
- No licensing
- No oversight committees

Key Insight: AI governance faces **multiple interconnected challenges!**

Power Asymmetries in AI Governance:

The Problem:

- Tech companies often more powerful than governments
- Wealth: Massive resources
- Influence: Lobbying and public relations
- Expertise: Advanced technical knowledge

Consequences:

1 Regulation Challenges:

- Governments struggle to keep up
- Companies resist regulation
- Asymmetric negotiation

2 Information Asymmetry:

- Companies know more
- Governments lack technical expertise
- Hard to regulate effectively

3 Accountability Gaps:

- Who holds companies accountable?
- Regulatory capture risk
- Limited enforcement

Key Insight: Tech companies may be **more powerful than some governments!**

Transparency Challenges in AI:

Why It's a Problem:

- Most AI systems developed by private companies
- Not subject to transparency requirements
- Limited information about:
 - Training datasets
 - Testing and tuning
 - Safety measures

Consequences:

1 No Public Oversight:

- Academics can't study systems
- Journalists can't investigate
- Public is in the dark

2 Regulatory Challenges:

- Regulators lack information
- Can't assess risks
- Enforcement difficulties

3 Trust Issues:

- Lack of transparency erodes trust
- Suspicion about practices
- Public concern

Key Insight: Private companies are **not subject to same transparency requirements!**

Interpretability Challenges:

The Black-Box Problem:

- Neural networks are "black boxes"
- Even experts can't fully understand
- Backpropagation doesn't provide insight
- Hard to trace decisions

Why It Matters:

1 Accountability:

- Can't hold accountable what you can't understand
- Responsibility gaps
- Who is responsible?

2 Auditing:

- Hard to audit black boxes
- New methods needed
- Limited effectiveness

3 Governance:

- Can't govern what you can't understand
- Regulatory challenges
- Oversight difficulties

Key Insight: Neural networks are **black boxes** - even experts can't fully understand them!

Proxy Variables and Governance:

What Are Proxies?

- Techniques to simplify training
- Represent complex objectives with simpler metrics
- Example: Word count as proxy for article quality

The Challenge:

1 Alignment Issues:

- Proxy may not align with true objective
- Optimization for wrong goal
- Unintended consequences

2 Opacity:

- If proxies are unknown
- Hard to understand model behavior
- Governance difficulties

3 Scale and Complexity:

- Billions of parameters
- Hard to trace processes
- Audit challenges

Key Insight: Unknown proxies make governance difficult!

Lack of AI Ethics Structures:

Comparison with Medical Ethics:

Medical Ethics	AI Ethics
Bioethicists	Few AI ethicists
Ethical codes	Developing codes
Ethics committees	Rare committees
Required coursework	Optional courses
Licensed professionals	No licensing
International codes	Fragmented guidance

Key Gap:

- Computer scientists can graduate without ethics training
- No certification or licensing
- Boards rarely include AI ethicists
- No oversight committees

Key Insight: AI ethics lacks the **established structures** of medical ethics!

The Regulatory Gap:

Current State:

- Few national laws (China exception)
- No specialized agencies
- No government-mandated auditing
- Voluntary frameworks only

Historical Parallel:

- FDA established for food and drugs
- Other regulatory agencies developed
- AI may follow similar path
- EU leading with AI Office

Why Regulation Matters:

- AI is international technology
- Cross-border data flows
- Need for cooperation
- International standards needed

Key Insight: Few national AI laws exist (except China) - no specialized agencies!

Unpredictability in AI Systems:

Emergent Behavior:

- Properties not explicitly coded
- Capabilities that surprise
- Mistakes that are unexpected
- Interactions that are unforeseen

Consequences:

1 Safety Risks:

- Unintended outcomes
- Harmful behavior
- System failures

2 Governance Challenges:

- Hard to regulate unpredictable systems
- Misuse difficult to prevent
- Unknown risks

3 Trust Issues:

- Surprises undermine trust
- Unpredictable = unreliable
- Adoption barriers

Key Insight: Emergent behavior = unexpected capabilities not explicitly coded!

LLMs and Truth-Tracking:

How LLMs Work:

- Statistical inferences and probabilistic guesses
- Designed for plausible responses
- Not based on truth or knowledge

Risks:

1 Misinformation:

- Plausible but false information
- Hard to detect
- Spread at scale

2 Physical Safety:

- Incorrect medical advice
- Dangerous recommendations
- Real-world harm

3 Legal Risks:

- Libel and defamation
- Copyright issues
- Speech-related risks

Key Insight: LLMs give **plausible, not necessarily truthful** responses!

Privacy and Copyright Challenges:

Privacy Concerns:

- Data scraped from internet
- Personal data of unsuspecting users
- GDPR compliance uncertain
- Right to access, modify, delete unclear

Copyright Concerns:

- LLMs ingest copyrighted material
- Books, articles, images
- Lawsuits in process
- Legal uncertainty

Governance Nightmare:

- Machine-created speech
- Company ownership
- Data acquisition unclear
- Regulatory challenges

Key Insight: Scraping data raises [privacy and copyright concerns](#)!

Global AI Regulatory Landscape:

Key Players:

1 European Union:

- GDPR (Privacy)
- DMA (Digital Markets)
- DSA (Digital Services)
- AI Act (Comprehensive AI Law)

2 United States:

- No federal privacy law
- State-level privacy laws
- AI Executive Order
- NIST AI RMF

3 China:

- Generative AI Measures
- PIPL (Privacy)
- Algorithmic Recommendation Rules

Key Insight: Different regions have **different regulatory approaches!**

European GDPR (General Data Protection Regulation):

Key Facts:

- Implemented May 2018
- Regulates personal data
- Applies to data controllers and processors
- Extraterritorial jurisdiction

Data Subject Rights:

- 1 Receive concise information
- 2 Access personal data
- 3 Request erasure
- 4 Object to processing

Obligations:

- Notify breaches within 72 hours
- "Legitimate interest" required
- Significant fines for non-compliance

Key Insight: GDPR = [European data protection regulation](#) with global reach!

The EU AI Act:

Key Facts:

- Approved March 2024
- Council approved May 2024
- Establishes common regulatory framework
- Risk-based approach

Risk Categories:

1 Unacceptable Risk (Banned):

- Biometric categorization using sensitive characteristics
- Untargeted facial image scraping
- Emotion recognition in workplace
- Social scoring
- Subliminal manipulation

2 High-Risk:

- Impact assessments required
- Thorough records
- Oversight measures

3 Limited/Minimal Risk:

- Transparency obligations
- Labeling requirements

Key Insight: EU AI Act = [first comprehensive AI regulation](#) - risk-based approach!

US AI Regulatory Framework:

Key Initiatives:

1 Executive Order on AI (2023):

- Safe, secure, trustworthy development
- Address AI-enabled harms
- International cooperation
- Safety testing requirements

2 NIST AI RMF:

- Voluntary framework
- Govern, map, measure, manage
- Flexible and adaptable
- Non-prescriptive

3 AI Safety and Security Board:

- Department of Homeland Security
- Critical infrastructure focus
- Recommendations and guidance

Key Insight: US has [executive orders](#) and [voluntary frameworks](#) - no federal AI law yet!

Comprehensive Summary: Responsible AI

Ethical Foundations:

- Consequentialism (outcomes and utility)
- Deontology (duties and rules)
- Virtue Ethics (character and virtues)

Core Principles:

- Nonmaleficence, beneficence, justice, autonomy, explainability

Key Practices:

- Bias identification and mitigation
- Privacy protection and data minimization
- Transparency and explainability
- Human oversight and accountability

Governance and Regulation:

- Ethical committees and structures
- Algorithmic auditing
- Compliance with regulations (GDPR, EU AI Act)
- Continuous monitoring and improvement

Exam Success: Understand **ethical frameworks, principles, governance, and regulation** for responsible AI!